# Payments & Refunds Functional Specification

## Authorization, capture, refund, idempotency and failure behavior

Document status: CURRENT unless otherwise identified | Enterprise delivery assurance training corpus

## 1. Scope and Terminology

Northstar Commerce is preparing a multi-release modernization of its digital commerce platform. This specification is intentionally detailed and includes overlapping business, API, testing, and governance content so that delivery-assurance systems must reason across multiple artifacts rather than rely on one isolated paragraph.

Payment authorization, capture, and refund are asynchronous in some downstream integrations. The functional specification defines stable business states separately from transient transport errors.

### 1.1 Operational Guidance

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# 2. Payment Authorization

Authorization requests must include a business idempotency key. Repeated submissions with the same key and same business intent should return the existing transaction rather than creating a second charge.

A gateway timeout does not prove that authorization failed. The platform must reconcile final payment state before allowing a second authorization attempt.

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# 3. Refund Processing

Refunds may be full or partial subject to remaining refundable amount. Duplicate refund submission must not exceed the eligible amount.

A refund may be accepted for asynchronous processing and later reach COMPLETED or FAILED.

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# 4. Error and Recovery Behavior

Transport errors, business declines, duplicate requests, validation failures, and dependency outages require different user and operational handling. Error mappings must remain stable enough to support automated tests.

| Risk | Expected Control | Release Impact |
| --- | --- | --- |
| P0 requirement has no active test | Coverage gap must be reviewed | Blocks release unless formally waived |
| Current and archived requirement conflict | Current approved version wins | Requires source/version evidence |
| Failed S1/S2 defect remains open | Release risk review required | Normally blocking |
| Requirement is ambiguous | Clarification required before oracle | Test generation should not invent rule |

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# 5. Audit Evidence

Payment and refund actions must retain correlation identifiers, actor context where applicable, amount, currency, decision, and final business state. Sensitive payment data must not be included in general application logs.

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